# Comprehensive Analysis of Customer Sales Data for the year 2025

## Brief Summary Report

Submitted by -
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# About the dataset

This summary report is prepared from a random dataset (Kaggle).

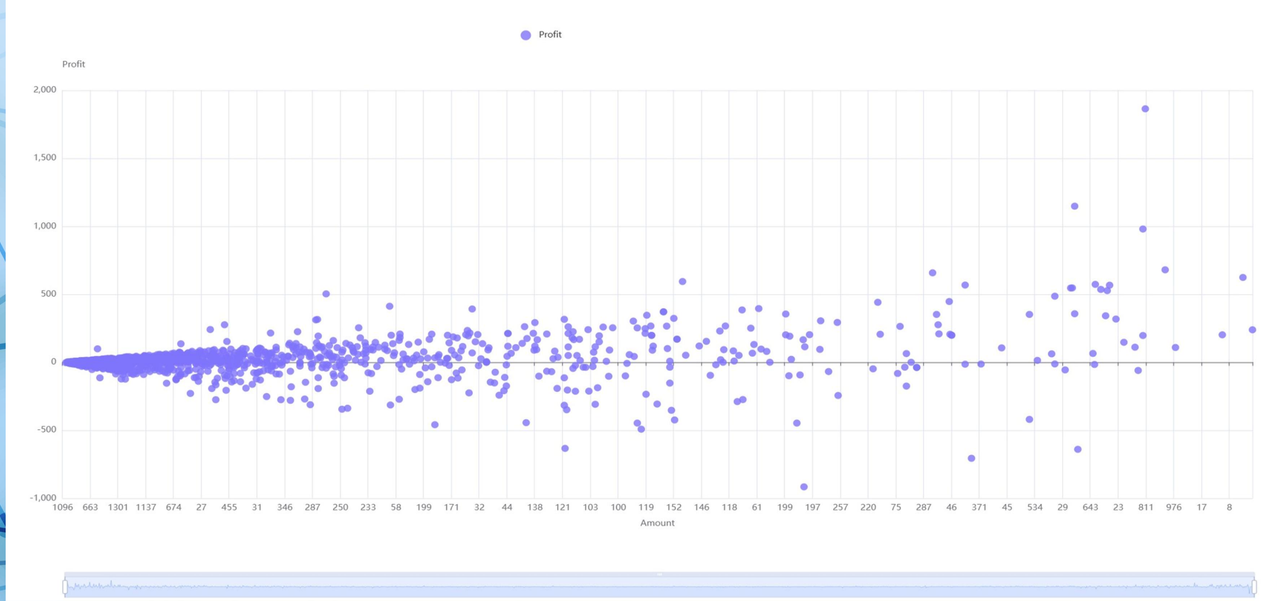
The dataset consists of ‘Two’ parts:
One with over 1500 transaction records detailing Order ID, Amount, Profit, Quantity, Category, Sub-Category and Payment Mode, useful for analyzing profitability and sales strategies.
Other part contains entries with Order ID, Order Date, Customer Name, & State aiding in understanding purchasing trends and customer preferences across different regions.

Together, these datasets support ‘Sales tactic refinement’, ‘Financial reporting’, ‘Sales forecasting’ and many more.

# The monthly sales trend for 2025

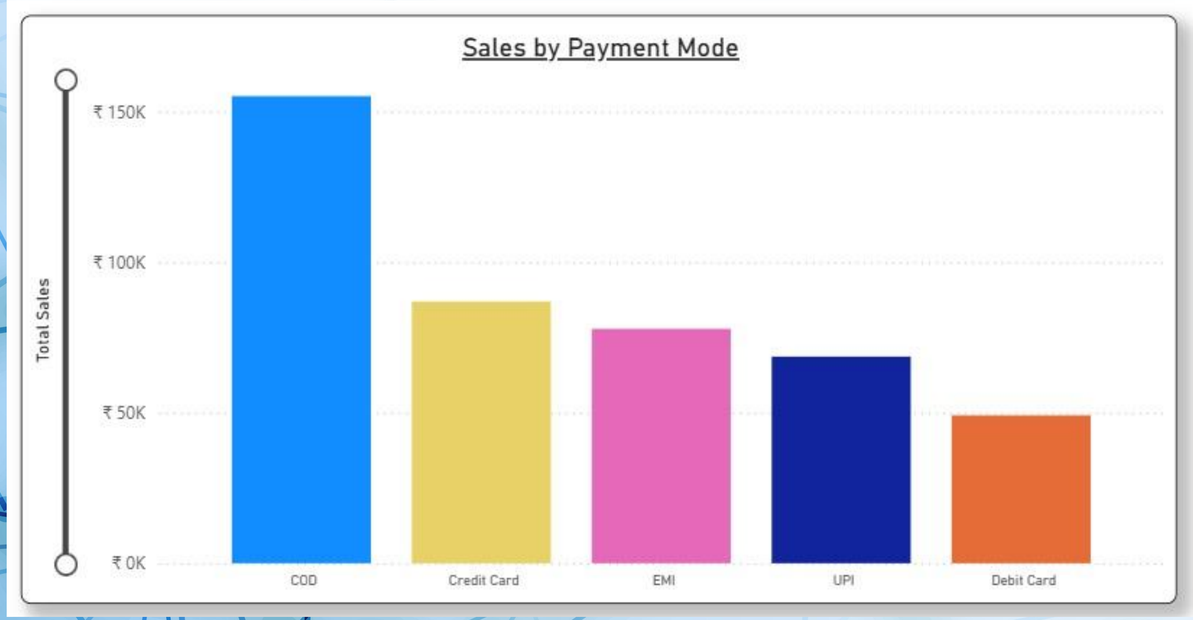
- **Sales Performance - January** had the **highest sales** at **61,632**, while **July** had the **lowest** at **12,966**. The **average monthly sales** were **36,480.92**.
- **Sales Trends -** Significant **fluctuations** with peaks in **March** and **November**. A **downward trend** was noted in **May** and **July**.
- **Strategic Insights -** Focus on boosting sales in **low-performing months** and analyzing high sales periods.

# Correlation between Sales amount and Profit



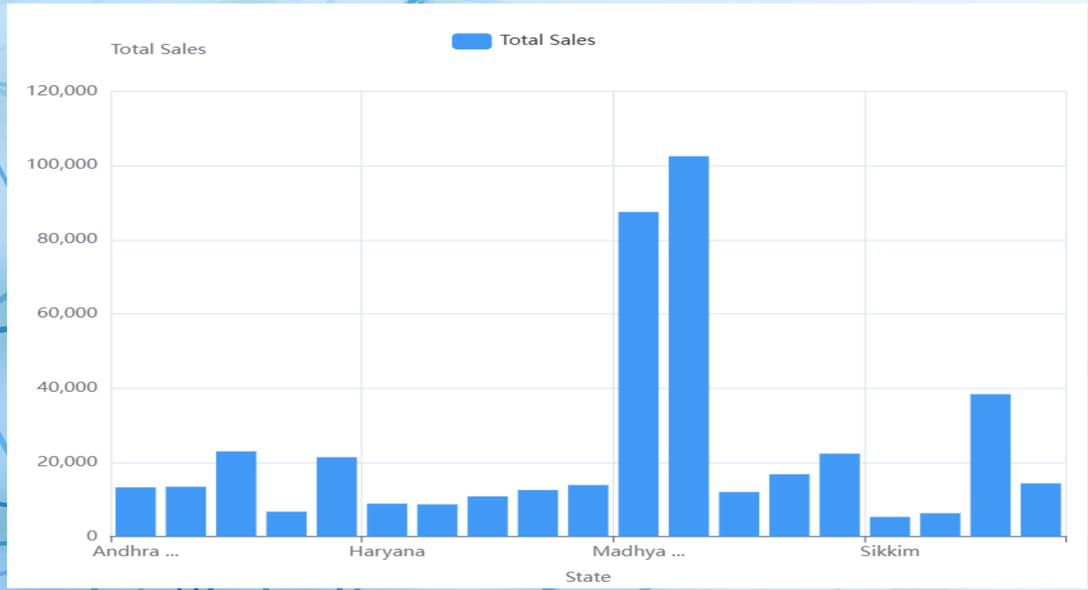

- **Moderate Positive Correlation** – The **correlation coefficient** of **0.31** indicates a moderate positive relationship between sales amount and profit.
- **Data Variability** - Average sales amount is **291.85** and profit is **24.64**, with significant variability, suggesting other influencing factors.
- **Scatter Plot** - Shows a slight upward trend with outliers, supporting the moderate correlation.

# The conversion rate by payment mode




- **COD** has the **highest transaction rate** at **1.368**, making it the most effective payment mode.
- **EMI** has the **lowest transaction rate** at **0.24**, indicating it is the least effective.
- **Conclusion -** Focus on enhancing **EMI** to improve its effectiveness.
- **Sales Data Analysis**: **COD** leads with **155,181**, while **Debit Card** is the least with **49,136**.
- **Visualization**: A **Bar Chart** highlights COD's dominance in sales.
- **Conclusion**: COD is the most popular payment mode; Debit Card is the least used.

# The sales distribution across different states

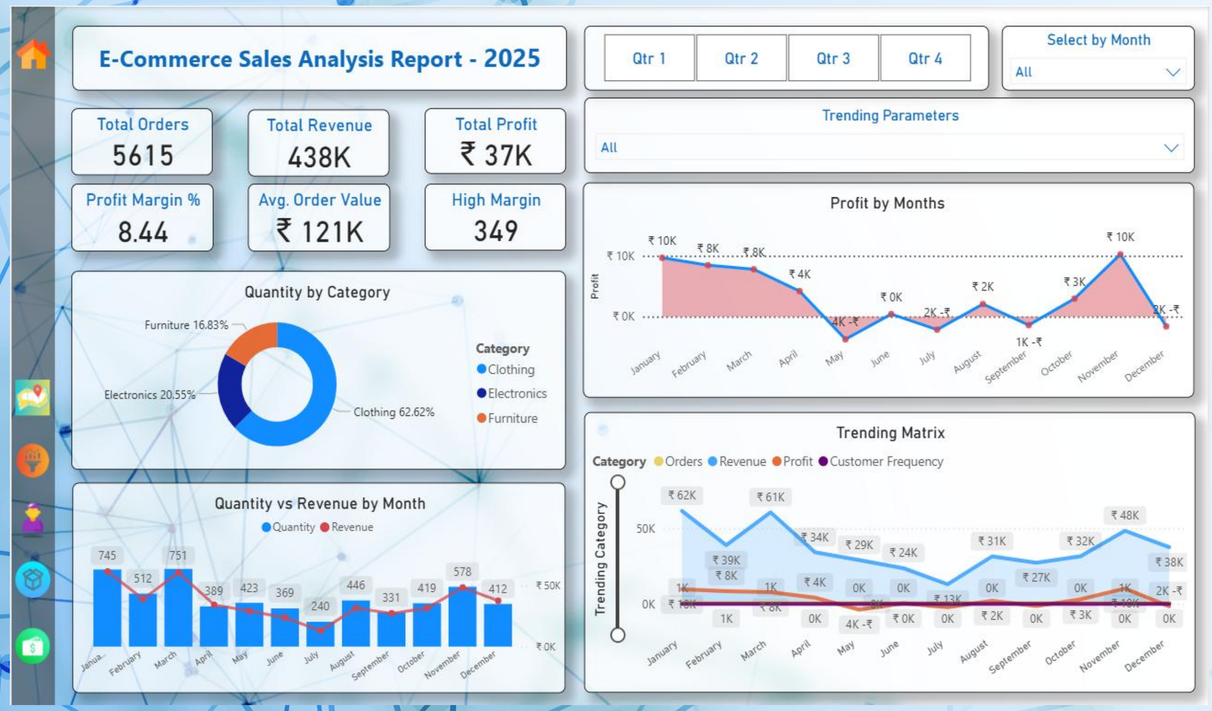



- **Sales Concentration** – **Maharashtra** and **Madhya Pradesh** lead in sales with **102,498** and **87,463** respectively.
- **Sales Variation** – Significant disparity in sales, with **Sikkim** at the lowest with **5,276**.
- **Growth Opportunities** – Potential for market expansion in states with lower sales like **Sikkim** and **Goa**.

# The sales contribution by different customer segments

- **Contribution** – Average segment contribution is **1.03%**; Maharashtra (Mumbai) leads with **4.08%**.
- **Key Contributors** – Maharashtra and Madhya Pradesh are significant, with Mumbai and Indore as top cities.
- **Visuals** – Bar chart highlights contributions from Maharashtra and Madhya Pradesh.

# The profit distribution by sub-category

- **Profit Analysis** – **Printers** lead with the highest profit of **8606**, while **Furnishings** have the lowest at **-806**. Sub-categories like **Electronic Games** and **Furnishings** show negative profits.
- **Visualization Insights** - Bar charts highlight **Printers** and **Bookcases** for high profits, and **Furnishings** for losses.
- **Conclusion** - Focus on profitable areas like **Printers** and **Bookcases**; address the losses in **Furnishings** and **Electronic Games**.

# Dashboard’s Screenshot